Fiscal Year:	FY 2022	**Task Last Updated:**	FY 07/12/2023
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Project Title:	Machine Learning as a Countermeasure to Spatial Disorientation in a Spaceflight Analog Environment		
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Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Go for Launch
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 04/30/2022 as per TRISH Science Office. Original end date was 11/30/2021. (Ed. 10/29/21)		
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Task Description:	Go for Launch Project Spatial disorientation may cause mission ending fatalities. We created a spaceflight analog condition where subjects stabilized themselves while spatially disoriented. We will use machine learning to predict destabilizing joystick deflections, loss of control, and crashes in this condition. We will also develop an optimal model controller that can suggest the best joystick deflection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our Research Impact includes: Deep Learning Study: - Shows, for the first time, that deep learning models can effectively predict crashes in a disorienting spaceflight analog balance task. - Shows how to modify deep learning techniques to use relatively small sample sizes which are uncommon in most deep learning implementation however which will be the case for spaceflight. - Shows how to create a general model that does not depend on many predefined parameters which allows programmers to make fewer assumption about the nuances of Mars or the Moon. - Shows how to train a deep learning model using non-optimal data with very unusual human behavior due to disorientation which will be the case during space exploration. - Highlights the future where human and artificial intelligence (AI) can learn together in a novel environment. Spatial Acuity Study: - Shows, for the first time, the huge individual differences in perception of self-orientation in a disorienting spaceflight analog condition. - Shows that there is no correlation between an individual's spatial acuity in the spaceflight condition and their ability to perform in it, thereby suggesting that training can overcome individual limitations in sensing orientation. - Shows that susceptibility to vestibular stimulation may predict early performance when disoriented. - Such vestibular stimulation may be a valuable addition to procedures for assessing individual differences in performance during initial exposure to a disorienting spaceflight analog condition where there are no reliable gravitational cues.
Task Progress:	Were astronauts forced to land on the surface of Mars using manual control of their vehicle, they would not have familiar gravitational cues because Mars' gravity is only 0.38g, and they could become susceptible to spatial disorientation, potentially causing mission ending crashes. In our earlier studies, we secured blindfolded subjects into a Multi-Axis Rotation System (MARS) device that was programmed to behave like an inverted pendulum. Subjects were instructed to use an attached joystick to stabilize around the balance point. We created the spaceflight analog condition by having subjects dynamically balance in the Horizontal Roll Plane, where they did not tilt relative to the gravitational vertical and therefore could not use gravitational cues to determine their position and had to rely only on motion cues. The crash boundaries were set at ±60 deg from the balance point and when it was reached the MARS would reset back to the balance point. We found 90% of subjects in our spaceflight analog condition reported spatial disorientation and all subjects showed it in their data. Compared to the control condition (Vertical Roll Plane, where gravitational cues from tilt were available), the rate of crashes was significantly higher. Our goal was to use deep learning to predict the occurrence of crashes before they happened. After trying a variety of recurrent neural networks (RNN) and non-RNN models, we chose the stacked gated recurrent units (GRU). We were able to predict crash events 800ms in advance with an AUC (area under the curve) value of 99%. We prioritized reducing false negatives (when the model incorrectly predicts that there would be no crash) and found it resulted in more false positives (when the model incorrectly predicts that there will be a crash). We found that false negatives occurred when subjects made unexpected destabilizing joystick deflections that rapidly moved the MARS away from the balance point. These unexpected destabilizing joystick deflections were likely a result of spatial disorientation and the reason why we could not predict crashes too far in advance. False positives often occurred at large values of displacement from the balance point and at high velocities, which could be used as a warning system in a future implementation. If our model could work in real time, we calculated that immediate action would result in the prevention of 80.7% of crashes however if we accounted for human reaction times (~400ms), only 30.3% of crashes could be prevented, suggesting that one solution could be an artificial intelligence (AI) taking momentary control of the spacecraft during these moments. No one to our knowledge has used deep learning to predict the occurrence of crashes in a novel analog condition where participants experience disorientation similar to what pilots and astronauts may experience. Space exploration will often demand astronauts to solve problems independently because of factors such as time delays in communication with Earth. Additionally, because astronauts will be exploring novel environments, optimal solutions to problems will not be known. These problems can arise both in the short term, such as spatial disorientation experienced during gravitational transitions and in the longer term such as effects to the brain from sustained long duration spaceflight. The consequences this will have on spaceflight are not fully understood especially when combined with multiple simultaneous stressors caused by factors such as radiation, psychological and physiological changes. Therefore, it is important to develop countermeasures such as artificial intelligence systems that are tested under a range of spaceflight analog conditions on Earth and can learn and adapt as astronauts collect data in space. In our approach, we develop an artificial intelligence system that does not have prior knowledge of the paradigm (such as inverted pendulum dynamics and the human vestibular system) and is not trained on optimal behavior (such as in the vertical roll plane where participants have task relevant gravitational cues). Instead, our model is trained on data obtained from our disorienting spaceflight analog task. Because our model did not rely on detailed knowledge of the paradigm it suggests that this methodology could be applied to other novel and disorienting conditions such as drifting during brownout, loss of control in human postural balancing in artificial gravity or from vestibular deficits. In a separate study, to determine acuity, blindfolded subjects experienced several patterns of passive movements while pressing a joystick trigger to indicate every time they passed the start point. We calculated spatial acuity by taking the standard deviation of the trigger presses. We correlated subjects' spatial acuity in the passive task to their ability in the active balancing task. We did not find any relation between Horizontal Roll Plane passive spatial accuracy and active balance control ability in the Horizontal Roll Plane, suggesting that spatial acuity and thresholds alone do not predict performance in a disorienting spaceflight analog condition. However, we did find significant correlations between performance in early active balancing in the Horizontal Roll Plane and spatial

	acuity in the Passive Vertical Roll Plane. These correlations were only significant for passive acuity trials conducted after vestibular stimulation was given that induced discomfort and slight motion sickness. Such vestibular stimulation may be a valuable addition to procedures for assessing individual differences in performance during initial exposure to a disorienting spaceflight analog condition where there are no reliable gravitational cues.
Bibliography Type:	Description: (Last Updated: 06/09/2025)
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